

2.1. Redstone's Problem

$$\min \mathbf{1}^\top G^2 \mathbf{z}$$

$$\text{s.t. } A\mathbf{x} \leq \mathbf{b}, \quad \mathbf{x} \in \{0, 1\}^N$$

$$\mathbf{z} \geq E\mathbf{x} + \mathbf{f} - \mathbf{1}, \quad \mathbf{z} \geq \mathbf{0}$$

Modeling Operation Cadence Constraint

Modeling Revisit Time Cost Function

2.2 Auxiliary Functions

- Adjust τ for each constraint element
- Adjust G for each cost function element
- Place multiple payloads in constraint object

3. SAT-to-GS

Constraint: SAT Ops Cadence

Cost Function: Each GS Revisit

4. GS-to-SAT

Constraint: GS Ops Cadence

Cost Function: Each SAT Revisit

3.1. Earth Observation SAT Image Capture Scheduling

- Get image data from ground by any available satellite
- Minimize revisit time for each ground operation point

3.2. 6G SAT Data Downlink/Uplink Scheduling (With ISL)

- Uplink/Downlink data packet from GS to any satellite
- Minimize time gap between each transmission
- Assume every satellite is connected each other by ISL

4.1. Distributed Ground Asset Scheduling for SSA

- Get SSA data of satellite by any available ground asset
- Minimize observation gap for each satellite

4.2. Earth Observation SAT Data Download Scheduling

- Download image data to any available ground station
- Evaluate data acquisition performance for each image